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GEOMETRIC ANALYSIS OF SHELL COILING: GENERAL PROBLEMS

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ABSTRACT—Among the shells of invertebrates that exhibit spiral growth, differences in form can be expressed by differences in geometric parameters. If three of these parameters are considered at a time, the spectrum of possible shell forms may be shown by a block diagram. Analog and digital computer constructions make it possible to visualize shell forms that are theoretically possible but do not occur in nature.

Actual species are not randomly distributed in the total spectrum of theoretically possible forms. Functional and evolutionary groups are confined to discrete regions of the spectum. Rational explanations for several of the observed distributions are apparent and may be expressed in terms of shell function. For example, a bivalve must have non-overlapping whorls in order to have a functional hinge. This fact restricts the geometric range of both brachiopods and bivalved molluscs.

Ontogenetic change in coiling geometry may be interpreted as a compensation for the effects of increase in absolute size during growth.

INTRODUCTION

THE shells of a wide variety of invertebrate animals have a coiled, or spiral, form. Examples may be found in many diverse groups: from the single-celled Foraminifera to highly complex groups such as the Mollusca and Brachiopoda. Coiling has developed independently in several quite distinct evolutionary lines. Its functional significance to the animal is not always the same, even within fairly homogeneous groups.

When one compares coiled shells from a variety of taxonomic groups, marked differences in form are evident. In spite of this variation, however, most types have important geometric characteristics in common which make rigorous comparison between them possible. The problem of defining parameters through which coiled forms can be analyzed has tempted many workers.

The literature of shell coining is long and varied. Perhaps the best known discussion is contained in D'Arcy Thompson's (1942) monumental volume "On Growth and Form." Thompson's treatment is essentially a review of earlier work published by others but it is valuable as a synthesis. More recently, work by Lison (1949), Owen (1953), Fukutomi (1953), Rudwick (1959), Stasek (1963), and others has added greatly to our knowledge of the coiling phenomenon.

As a result of these studies, there is general agreement that the bulk of the coiled forms observed in nature are but variations on one rather simple model. Several sets of parameters have been suggested to describe differences and similarities between coiled forms. But there have been few if any successful applications of this knowledge to the specific problems facing the zoologist or paleozoologist working with animals having a coiled shell. From the standpoint of application, the analysis of shell coiling has been little more than intellectual exercise. Its practical importance has never been established.

There are many reasons for this: I have discussed several of them in an earlier paper (Raup, 1961), but the most important is that the basic relationships between shell geometry and shell function have not been clearly presented. The purpose of the present paper is to explore some of these basic relationships. The discussion will consider broad problems of the distribution of functional and taxonomic groups within the spectrum of geometrically possible forms. A subsequent paper (Raup, in press) will be a detailed analysis of coiling variation in the Ammonoidea.

Both papers are presented not only as an attempt to broaden the applicability of coiling analysis but also as an example of the methods of "theoretical morphology" (Raup & Michelson, 1965). Using this approach, a conceptual or mathematical model is established for some aspect of morphology. The model in turn is used to formulate or describe the total spectrum of physically possible forms. This spectrum then serves as a framework through which actually occurring forms may be interpreted.

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GEOMETRIC MODELS FOR COILING

Moseley (1838) established a geometric model for shell coiling based on the logarithmic spiral. This model has been accepted by most zoologists and paleozoologists and is widely known through Thompson's review (1942). Moseley based his analysis primarily on gastropods but clearly intended that his model be applicable to essentially all coiled forms—both planispiral and helicoid.

Two years after Moseley's initial publication, Naumann (1840a) published a similar analysis but based his model on a more elaborate spiral. He argued that Moseley's logarithmic spiral did not provide a reasonable approximation of actual morphology. Considerable discussion in the literature followed (Moseley, 1842; Naumann, 1840b, 1845, 1848). Naumann's work was carried on and expanded by Müller (1850, 1853) and Grabau (1872). In the latter works, most of the species used were ammonoids and therein lies the reason for the differing interpretations of Naumann and Moseley. Ammonoids often show considerable ontogenetic change in geometry, particularly in whorl shape and degree of involution. Thus, it is commonly impossible to find a logarithmic spiral that fits the shell throughout ontogeny. Much of the work of Naumann and his students was an effort to produce an equation or set of equations that would satisfy the ammonoid shell throughout ontogeny. Moseley, on the other hand, considering a wider variety of shell form and emphasizing gastropods (where ontogenetic variation, though present, is minimal), found the logarithmic spiral to give the most consistent accuracy. Moseley's case was strengthened by the fact that he was able to present *a priori* justification for his model: that the logarithmic spiral provides for growth without change in shape. This constancy of shape is obviously of functional importance to the majority of coiled forms—where mode of life and shell functions do not change during post-metamorphic growth. Thompson (1942) provided many other examples of good conformity to the logarithmic model, drawn from a wide variety of taxonomic and functional types, and he thus further confirmed the general applicability of Moseley's model.

In view of the foregoing, the logarithmic spiral is used as a basis for the present analysis. This does not, however, relieve us of the problem of ontogenetic variation in ammonoids and other groups that display it. Rather, it puts this ontogenetic variation in another light. Change in geometry during growth may be looked upon as a genetically controlled departure from the norm, and as such must be interpreted in terms of the functional or adaptive advantage of the departure to the organism. This subject will be considered at length below and in another paper (Raup, in press).

The logarithmic model used here is that of Raup & Michelson (1965). Four basic parameters describe the general form of the coiled shell. They are: the shape of the generating curve (S), the whorl expansion rate (W), the position of the generating curve in relation to the axis (D), and the rate of whorl translation (T).

The generating curve is often equivalent to the outline of the growing edge of the shell. As such, its shape can only rarely be defined mathematically. An equation can be used when the generating curve approximates a simple geometric figure such as a circle or ellipse. More generally, however, a line drawing is used to describe the shape. The other three parameters can be defined numerically.

The coiled shell may be considered in a system of cylindrical coordinates (text-fig. 1). The y-axis coincides with the axis of coiling, θ expresses revolution of the generating curve about the axis, and r is the distance of any point from the axis. The *initial* generating curve may be defined conveniently by a set of arbitrary points (such as point A in text-fig. 1) and is so placed in the coordinate system that the geometric center (point B in text-fig. 1) has a y-value of zero.

If r_o is the initial distance of point A (or any other such point) from the aixs then r_θ, the distance after θ revolutions, may be expressed as follows:

$$r_\theta = r_o W^{\theta/2\pi} \tag{1}$$

where:

W is the rate of whorl expansion.

In planispiral shells ($T=0$) the y-value of the same point is expressed by:

$$y_\theta = y_o W^{\theta/2\pi} \tag{2}$$

where:

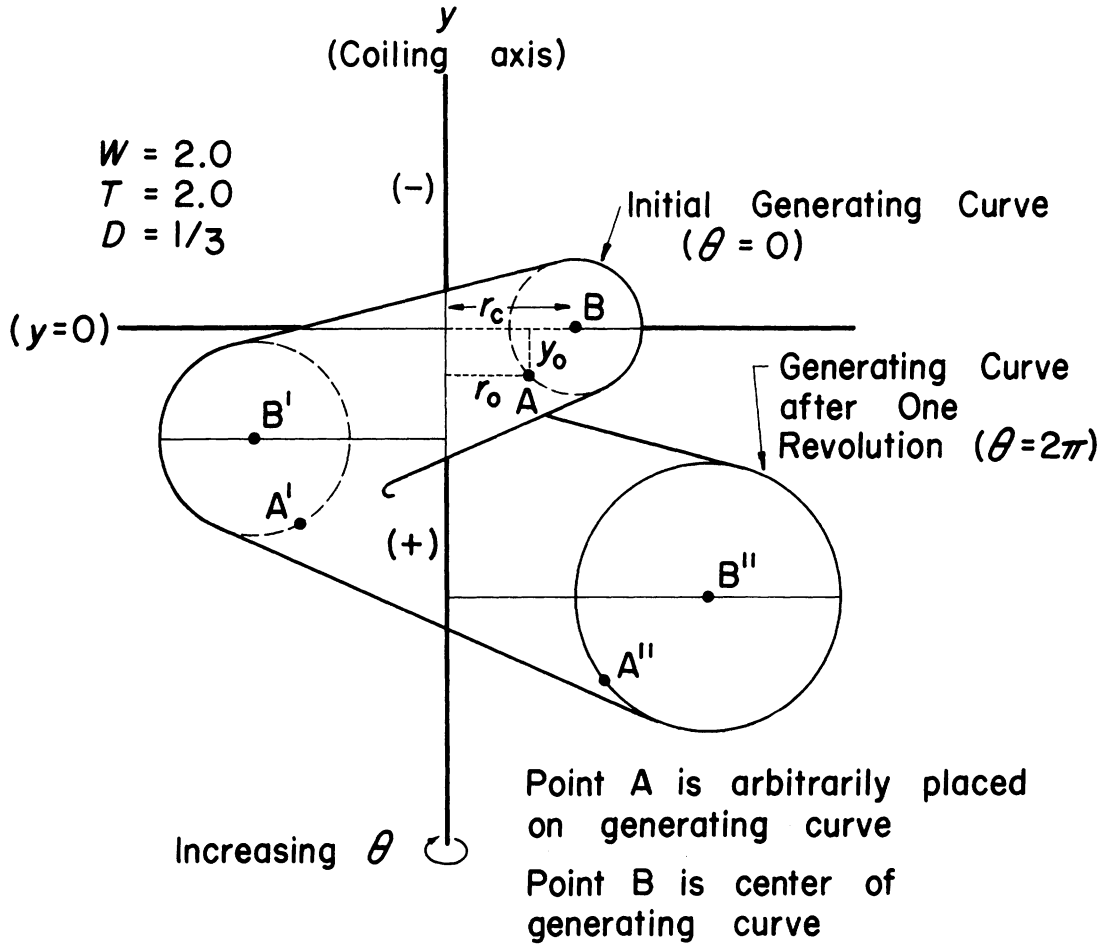
y_o is the initial y value of the point.

In planispiral forms, the geometric center of the generating curve has a constant y-value throughout growth ($y=0$). In helicoid forms, on the other hand, the generating curve moves along the y-axis and equation (2) becomes:

$$y_\theta = y_o W^{\theta/2\pi} + r_c T(W^{\theta/2\pi} - 1) \tag{3}$$

where:

T is the rate of translation defined as dy/dr, with respect to the center of the generating curve, and r_c is the r-value of the center of the initial generating curve.



Point	Coordinate		
	θ	r	y
A	0	r_o	y_o
A'	π	$r_o W^{1/2}$	$y_o W^{1/2} + r_c T(W^{1/2} - 1)$
A''	2π	$r_o W$	$y_o W + r_c T(W - 1)$
B	0	r_c	0
B'	π	$r_c W^{1/2}$	$r_c T(W^{1/2} - 1)$
B''	2π	$r_c W$	$r_c T(W - 1)$

TEXT-FIG. *1*—One whorl of a hypothetical coiled shell in cylindrical coordinates.

Equations (1) and (3) can be used to locate any point on the generating curve where a known angular distance of growth separates it from the initial generating curve. These equations have been used extensively in programming digital computers for graphical construction of ideal shell forms (Raup, 1962, 1963; Raup & Michelson, 1965). In these programs, W and T are input constants for a given shell form. The shape of the generating curve (S) is defined by the coordinates of points on the initial generating curve. The coordinates of these points also fix the position (D) of the curve relative to the coiling axis. When dealing with simple, ideal generating

curve shapes, as in text-fig. 1, a distance (D) is defined as the ratio between the r-value of the axial margin of the generating curve and that of the outer margin. D is thus zero when the generating curve is in contact with the coiling axis (as in many gastropods) and increases as the generating curve moves away from the axis (as in serpenticone ammonoids).

It should be emphasized that the four parameters (S, W, D, and T) do not completely describe the morphology of the coiled shell. The complete morphological picture would include the many features of surface ornamentation, growth rings, internal structure, shell thickness, and so on. The model is concerned *only* with general form and includes only that ornamentation produced by the revolution of the generating curve about the axis. Furthermore, the model does not describe ontogenetic change in coiling, though this can sometimes be added as an elaboration of the basic equations. The model assumes dextral and orthostrophic coiling. Hyperstrophic coiling can be produced by assigning a negative sign to T.

All four parameters can be measured readily in nearly all specimen material, as long as the position of the coiling axis can be located. The shape of the generating curve is equivalent to the shape of the whorl cross-section. The position of the generating curve (including its distance from the axis) may be observed directly or from a suitably oriented photograph. Translation rate is equal to the calculated ratio of movement of the center of the generating curve along the coiling axis to movement away from the coiling axis. The expansion rate may be calculated from measurements of one linear dimension on any two replicas (growth stages) of the generating curve, as long as the angular distance between the two is known (the difference between the two θ's). An accurate estimate of expansion rate may be made even where less than a full whorl is observed (as in many bivalves). In the calculation of expansion rate, any external dimensions of the whorl or distances between prominent features of spiral ornamentation may be used.

THE SPECTRUM OF POSSIBLE FORMS

As pointed out in an earlier paper (Raup & Michelson, 1965) the geometric model and the computer constructions based on it can be used to visualize the total spectrum of geometrically possible shell forms. This spectrum takes the form of a four-dimensional space with one dimension for each of the basic parameters. It is assumed that each parameter is independent of the others and that all combinations of the four variables are possible. One of the purposes of the present paper is to show how this hypothetical four-dimensional space is filled with actual species, living and fossil.

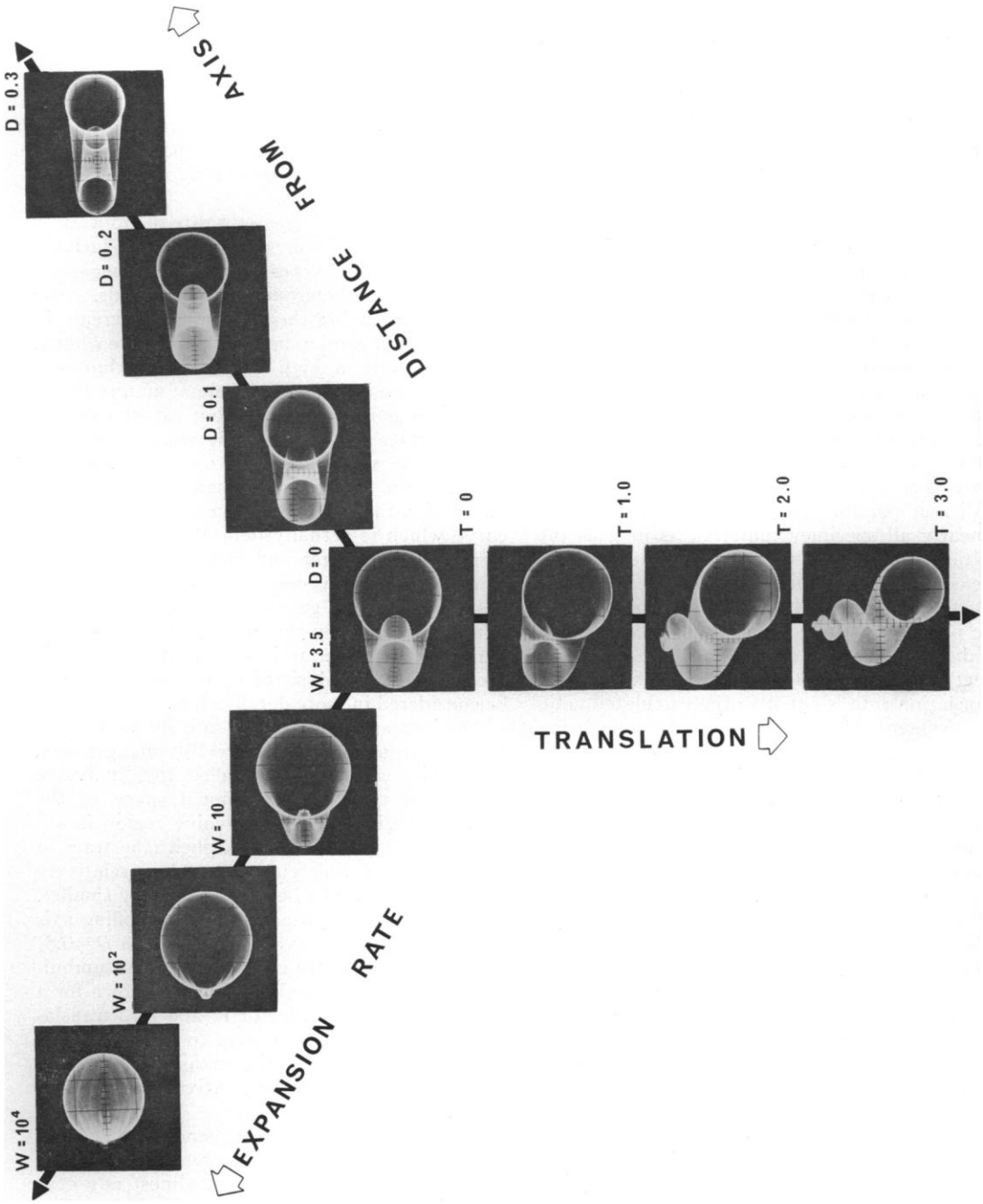
It is important, first, however, to have a clear picture of the effects on general morphological aspect of a change in each of the parameters. Text-figure 2 shows three series of hypothetical shell forms, each series starting from the same form. The initial form was chosen arbitrarily and is not intended to represent any particular taxonomic group. Nevertheless, it is a form commonly found in several groups.

In text-figure 2, increase in rate of translation (from $T=0$ to $T=3$) produces the relatively high-spired coiling types most common in gastropods and some heteromorph ammonoids. Note that accompanying this change is a decrease in the amount of overlap between successive whorls. In forms with a high degree of overlap, the organism does not deposit skeletal material over the entire generating curve, but rather uses the outer surface of the preceding whorl to enclose part of the new whorl. As overlap decreases, however (in this case through an increase in translation rate), that portion of the generating curve which is actually deposited by the organism becomes larger and larger. Only in cases where overlap of successive whorls is completely absent is the total generating curve, as a closed figure, deposited by the organism. The effect of variable whorl overlap is an important factor in the functional analysis of coiled forms and will be considered in more detail below.

The increase in expansion rate shown in text-figure 2 (from $W=3.5$ to $W=10^4$) changes shell form from what may be called the "univalve region" of the four-dimensional space to the "bivalve region". As the bivalve region is approached, the opening of the shell (the trace of the last generating curve) becomes relatively larger and the umbo becomes relatively smaller.

Increase in the distance from the coiling axis to the generating curve (from $D=0$ to $D=0.3$) makes the shell more evolute and more umbilicate. In text-figure 2 this change is shown for a shell having a low value of W and zero translation. If the same change were to take place in a form having a high W the morphological effect would be to increase the relative length of the umbo.

Change in the shape of the generating curve is illustrated in text-figure 3. Depending on the means used to describe shape, almost any configuration of the generating curve is possible. The apparently simple parameter S could be subdivided into several parameters, each adding dimensions to the four-dimensional space. In text-figure 3, the illustration in the center has a circular generating curve. The upper and lower

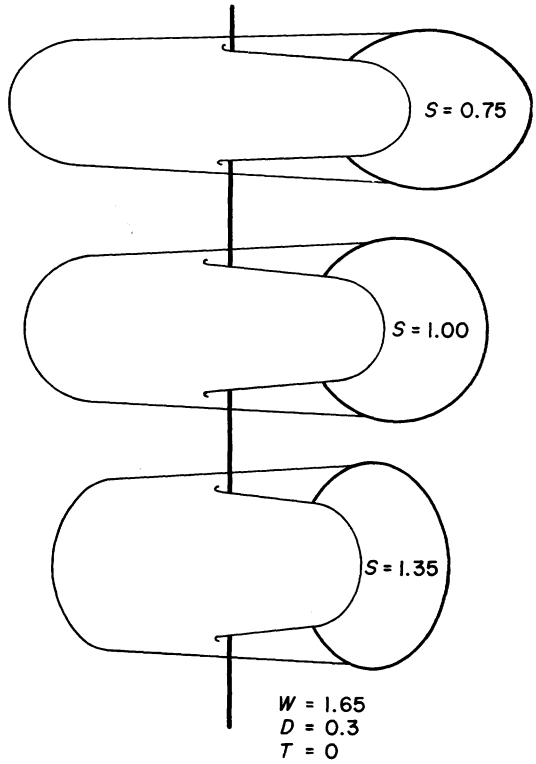


TEXT-FIG. 2—Computer produced shell forms illustrating the morphological effects of change in whorl expansion rate (*W*), translation rate (*T*), and the distance from the generating curve to the axis (*D*),

illustrations employ differently oriented ellipses. In this figure, S is defined as the ratio between axes of the generating curve parallel and normal to the coiling axis. Clearly this is only one kind of change in the shape of the generating curve which could be represented. This particular mode of change in generating curve shape is used in text-figure 3 because of its applicability to ammonoid geometry.

In text-figure 4 other possible combinations between the basic parameters are shown. Here the shape of the generating curve is held constant (a simple circle) and a *three*-dimensional space of geometrically possible forms is defined. The front vertical face of the block, the top face, and the interior of the block are occupied by helicoid forms (T greater than 0). The right hand vertical face contains planispiral forms ($T=0$).

Comparable block diagrams could be constructed for other shapes of the generating curve. Changing generating curve shape would, of course, change the morphological aspect of the resulting shell forms but the general relations shown in text-figure 4 would not change appreciably. For example, the form shown in the upper left would remain gastropod in aspect regardless of the shape of the generating curve. We can, therefore, use text-figure 4 as a format for analysis of the distribution of real world geometric forms if it is kept in mind that elaboration of the generating curve would complicate the picture.



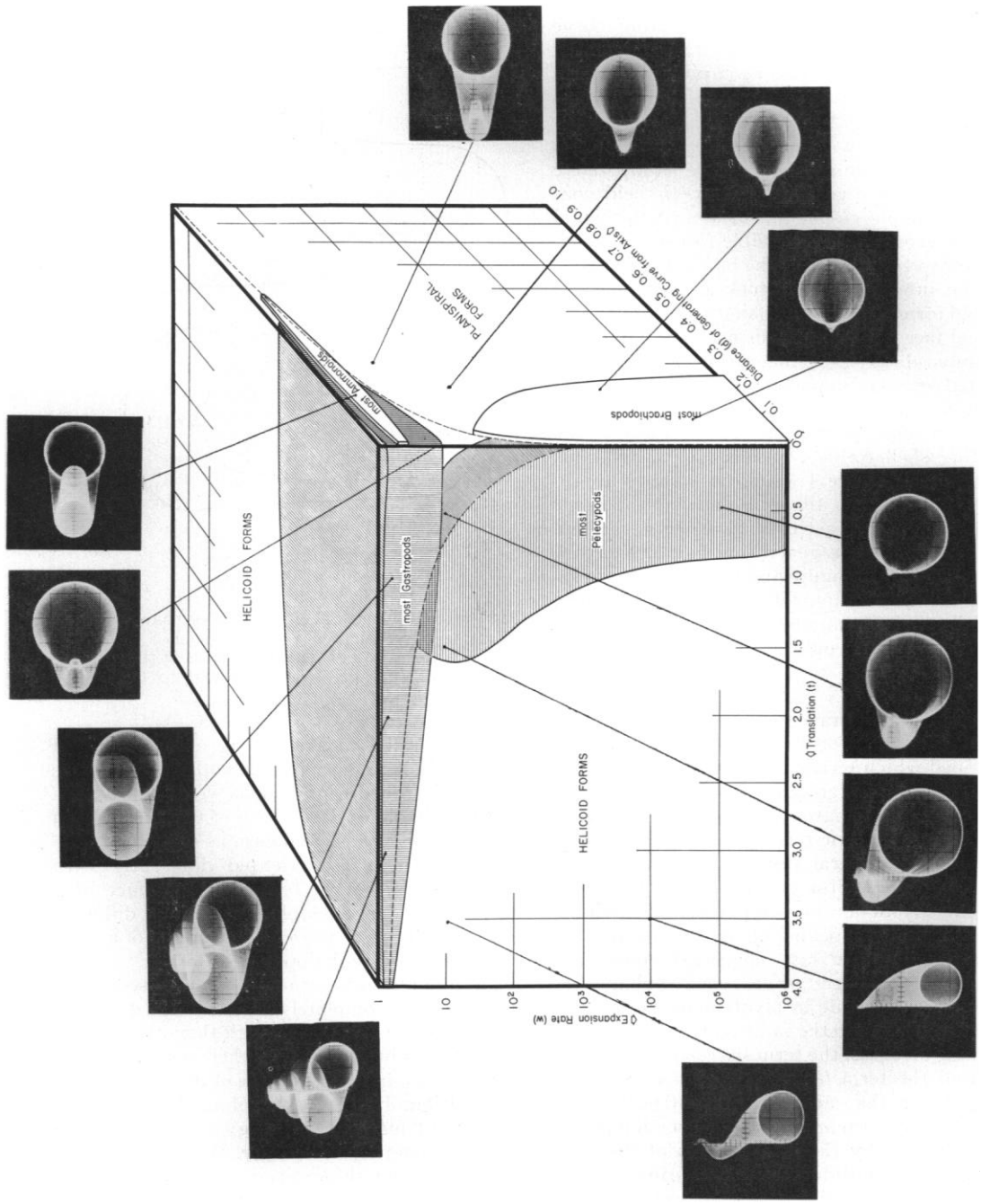
TEXT-FIG. 3—Morphological effect of change in generating curve shape. The parameter S is defined here as the ratio between dimensions of the generating curve (parallel to and normal to the coiling axis).

PROBLEMS OF TERMINOLOGY

For each of the taxonomic groups being considered here, a morphologic terminology has developed which serves the needs of that group. When the several groups are brought together under a single format as in text-figure 4, we find that the morphologic terminology often leads to confusion. In several instances one geometric element of the shell is given different names in different taxonomic or morphologic situations. For example, the opening of the shell (generally the outline of the last deposited generating curve) is termed *aperture* in coiled cephalopods and gastropods while in bivalves the term *commissure* is applied to the same feature.

Among bivalves, the term *interarea* in brachiopods and the term *ligament area* in pelecypods both refer to the same gemetric structure. In both cases the term applies to that portion of the shell deposited by the axial margin of the generating curve. Consider also the differing ways in which such simple dimensions as *length* are defined. In brachiopods the length dimension is generally defined as the maximum dimension through the beak and perpendicular to the coiling axis. In pelecypods, on the other hand, the length dimension runs approximately but not exactly parallel to the coiling axis. Length in gastropod shells also parallels the coiling axis whereas in coiled cephalopods we do not have a comparable dimension. Much of the variation in the definiton of basic dimensions stems from the fact that shells are oriented differently with respect to geometry. The ventral surface of a brachiopod, for example, is totally different geometrically from the ventral surface of a pelecypod or coiled cephalopod.

The differences in terminology discussed above are, of course, completely legitimate in terms of the taxonomic groups to which they correspond. Most of the differences have their roots in fundamental anatomical differences or in differences in life position. It would, therefore, be unwise to attempt to make terminology uniform for the sake of geometric consistency. We can, however, superimpose on these terminologies a set of informal geometric terms which will have meaning in the context of the present analysis. For example, when referring to a portion of the last deposited generating curve (aperture or commissure) the term *axial margin* has been used. The axial margin of the last generating curve in



TEXT-FIG. 4—Three dimensional block illustrating the spectrum of possible shell forms. The shape of the generating curve is assumed to be constant. The regions occupied by the majority of species in four taxonomic groups are outlined in the block. Species of these groups are not commonly found in the blank regions of the block.

brachiopod terminology would be the posterior margin of the commissure. In pelecypods it would be the dorsal margin of the commissure. In gastropods it would be the inner margin of the aperture and in cephalopods it would correspond to the dorsum.

NATURAL OCCURRENCE OF COILING TYPES

The foregoing discussion has concentrated on defining the range of possible shell forms from a strictly geometric viewpoint. To reduce the scope to a manageable scale certain ideal conditions have been assumed, as follows: a simple and constant generating curve shape, perfect adherence to the model (that is, no ontogenetic variation), and orthostrophic, dextral coiling. There is probably no group of coiled organisms that fulfills all these conditions perfectly. However, the approximation in nature is generally close and we can proceed to estimate what parts of the total spectrum of forms is or has been used by actual organisms.

An attempt has been made in text-figure 4 to outline the regions in which representatives of four major taxonomic groups are concentrated: Brachiopoda, Pelecypoda, Gastropoda, and Ammonoidea. The regions are not intended to encompass all occurrences for a given group. As an example, the ammonoid region is shown as being confined to a small part of the planispiral face of the block (including values of W from approximately 1.25 to 3.0 and of D from 0 to 0.65). These limits include perhaps 90% of all ammonoids: many of the remaining 10% fall in the interior of the block (turriculate forms).

Of the taxonomic regions shown in text-figure 4, only that of the ammonoids is actually based on measured data (Raup, in press). The regions for brachiopods, gastropods, and pelecypods are estimated from comparisons of computer-produced ideal forms and a representative array of species. In spite of the tentative nature of these regions, however, they are probably substantially correct. It cannot be questioned, for example, that brachiopods predominate in the high W, low D portion of the planispiral face. Similarly, the W values for pelecypods are on the average higher than for gastropods.

Inspection of text-figure 4 raises several questions. It should be noted, for example, that the bulk of the species in the four taxonomic groups are confined to *non-overlapping* regions which if taken together comprise a relatively small part of the block. Clearly, the distribution of actual species is not random. Do the relatively unused regions represent physiologically impossible shell forms or has the evolution of these taxa simply not had sufficient time in which to populate the entire block? Are some regions of the block suited only to one set of shell functions? Are swimming forms, for example, necessarily limited to certain discrete regions? The answers to these and similar questions are necessary if we are to understand fully the evolution of coiled organisms.

Text-figure 4 is representative of a wide variety of diagrams which could be drawn on the format of the three-dimensional block. The taxonomic range could be expanded, particularly to include foraminifera, nautiloids, and other groups of organisms that display coiled morphology. Going to lower taxonomic levels and thereby subdividing major taxonomic groups would also be meaningful. Such diagrams could be used to investigate the evolution of specific genera and families. From the point of view of shell function, distributions in this format could be shown of burrowing bivalves as opposed to epifaunal bivalves, swimming forms as opposed to sessile forms, etc.

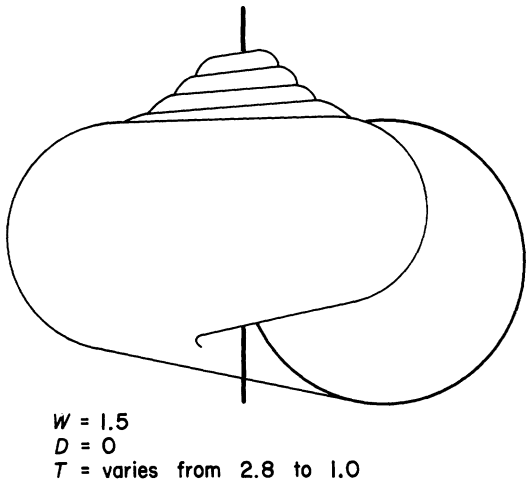
ONTOGENETIC CHANGE IN COILING GEOMETRY

In the foregoing discussion the assumption has been made that the geometric characteristics of a shell are constant throughout ontogeny. It was pointed out earlier, however, that this is not always the case. In geometric terms, the effect of ontogenetic change is to shift the organism from one part of the geometric spectrum to another. Specific cases of ontogenetic variation in coiling fall principally into two categories: either the ontogenetic change is sudden (as in vermetid gastropods and in some heteromorph ammonoids) or the ontogenetic change is gradual, so gradual that some authors have argued that the logarithmic model does not apply.

In examples of the first kind of ontogenetic change it can often be shown that the change coincides with a major change in mode of life (such as from planktonic to benthonic). In view of the functional importance of shell form, it is not unreasonable that a change in mode of life during ontogeny should be coincident with a change in geometry.

Text-figure 5 shows an example of gradual ontogenetic change. The example is purely hypothetical (produced by computer) but it is of a type commonly found among gastropods. The ontogenetic change involves a gradual decrease in translation rate (T) from 2.8 to 1.0. All other parameters are held constant ($W=1.5$, $D=0.0$, and the generating curve is circular). The morphological result is an "extraconical" shell form: whorl overlap increases steadily and the spire of the shell is concave in lateral profile.

The amount of ontogenetic change shown by the example in text-figure 5 is considerable when compared with the total range in T exhibited by the gastropods (text-fig. 4). Cases such as this



TEXT-FIG. 5—Hypothetical shell form showing the morphological effect of ontogenetic change in translation rate.

are uncommon but they do exist (and may even dominate certain genera and families). Quite a high degree of plasticity in geometry during growth is thus possible. It is important to note, however, that the plasticity is limited in kind. In text-figure 5, shell geometry has shifted from one part of the gastropod region (text-fig. 4) to another rather than across a boundary between major taxonomic regions.

In instances where ontogenetic change is gradual, many authors have suggested that the change is simply a relic of phylogeny. In certain cases, this may indeed be true. However, in the general case it seems more reasonable that the organism is adjusting during its growth to changes in the dynamics of shell function that are produced by growth itself. Such changes refer primarily to the effects of absolute size. Consider, for example, the problem of the pelecypod for which it is advantageous to maintain a constant relationship between the strength of its adductor musculature and the total mass of the shell. As the shell grows, following the ideal logarithmic model, the volume of the shell increases more rapidly than the area of the shell opening. (Volume increases as the cube of a linear dimension of the generating curve, whereas the area of the opening increases only as the square.) It would be entirely plausible, therefore, for the bivalve to alter its geometry during ontogeny in order to maintain a favorable ratio between area of the opening and volume of the shell.

Other effects of increase in absolute size include the strength of the shell wall in relation to shell size, and the amount of shell building material necessary to create a desired volume of shell interior. At this time, information is available only in a very few cases to interpret ontogenetic change rigorously in terms of compensation for increasing absolute size. Some of these will be considered later (Raup, in press). Other examples must wait for more detailed research.

THE PROBLEMS OF BIVALVENESS

In text-figure 4, the bivalve groups (pelecypods and brachiopods) have higher values of W than the univalves (gastropods and ammonoids). In fact the only region of overlap between these two major shell types is that small region in the neighborhood of $W=10$, $T=1$. This raises the question of what geometric characteristics a shell must have in order to operate as a bivalve.

It was noted earlier that change in one or more basic parameters may change the amount of overlap between successive whorls. If other factors are held constant, an increase in translation rate decreases the whorl overlap. Similarly, increases in expansion rate or in the distance between the generating curve and the axis both decrease whorl overlap. In a bivalve shell the hinge line is most important. In order for the two valves to articulate, the axial margin of the generating curve must be exposed. As was shown earlier, a form which has whorl overlap does not expose or even precipitate the axial margin of the generating curve. Thus, we can postulate that *in order for a geometric form to operate as a equivalved bivalve, whorl overlap must be absent.*

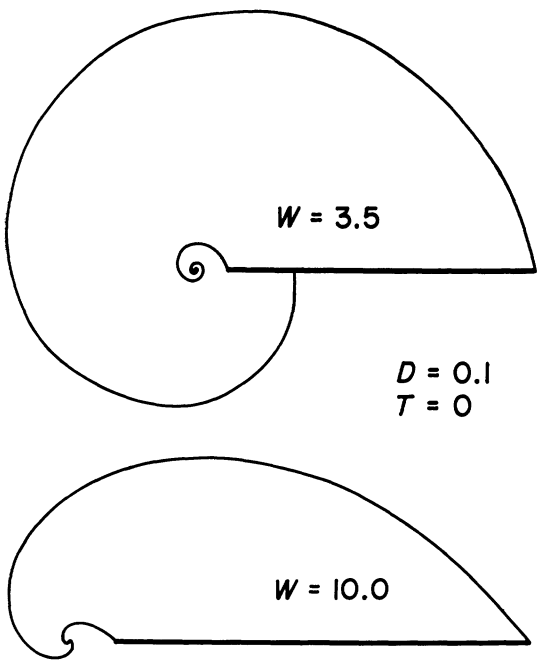
Text-figure 6 shows two idealized shell forms. The upper one shows overlap of successive whorls whereas the lower has a geometry such that the whorls are completely separate. Were the former to function as a bivalve, the hinge and dentition structures would of necessity be deposited as a secondary secretion of the shell on the outside of the preceding whorl, and would have to shift position constantly during growth. There would be no space for the development of a ligament area or interarea. From this viewpoint, the lower illustration in text-figure 6 is readily adaptable to the bivalve form.

We can now return to the theoretical consideration of determining what part of the total spectrum of geometrically possible forms contains shells which lack whorl overlap and thus could be bivalves. If we assume a generating curve of constant shape, it is possible to calculate rigorously a surface in the three-dimensional block which will separate forms where whorls overlap from forms where there is no overlap. Along the surface itself, successive whorls are just in contact. This surface is sketched in text-figure 7.

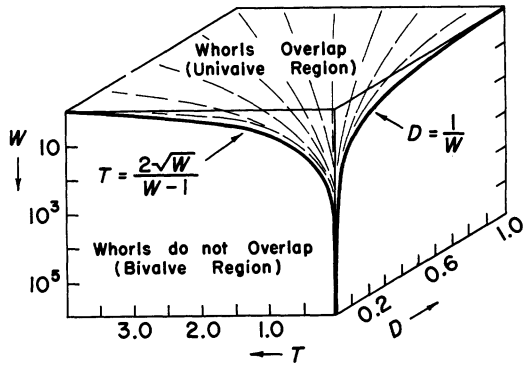
The upper form in text-figure 6 is derived from above this surface and the lower one from below the surface.

We can thus define the bivalve region as that region of the three-dimensional block lying below the surface just described. Note, however, that *bivalves do not occupy the entire region below the surface.* If the only requirements were those discussed above, the bivalves could occupy a larger part of the geometric range.

We may ask what is wrong from the viewpoint of the bivalve with the left side of the front helicoid face in text-figure 4. A geometric form in this region has a considerably higher internal volume than the typical bivalve. Compare, for example, the illustration in text-figure 4 on the front face at $T=3.5$ and $W=10$ with that at $T=.5$ and $W=10^5$. The relative size of the opening in the latter case is much greater. One of the primary functions of the opening in the bivalve shell is to provide space for adductor muscles passing from one valve to the other. Furthermore, the required strength of the musculature must be proportional to the mass or volume of the whole shell. Strength of muscles is in turn proportional to their cross sectional areas. It is possible to imagine, therefore, a shell which would have too small an opening in relation to its total mass to accommodate the musculature



TEXT-FIG. 6—Hypothetical shell forms illustrating the effect of whorl expansion rate on whorl overlap. The geometry of the upper form is typically univalve and that of the lower is typically bivalve.

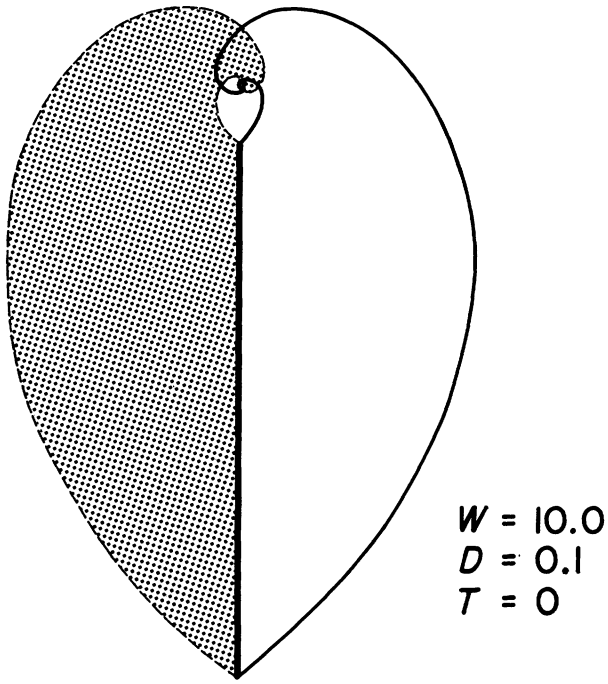


TEXT-FIG. 7—Theoretical surface separating the univalve region from the bivalve region (format is that of text-fig. 4).

necessary for valve opening and the same time provide space for other vital organs. As we move from left to right on the front helicoid face of the block we are moving geometrically from a region of high ratio between internal volume and the area of the opening, to a region having a more favorable ratio from the viewpoint of these relationships. It is probably for this reason that we find the pelecypods confined mostly to the region of relatively low translation. By a similar argument the brachiopods are plausibly confined to the left hand portion of the planispiral face of the block.

Organisms using the bivalve shell form face other critical problems. Some of these have been discussed by Stasek (1963). Text-figure 8 shows a hypothetical shell identical to that in the lower of the two illustrations in text-figure 6. If we add to this shell its mirror image (dashed in text-figure 8), such that the two "shells" meet along the generating curve, we see that even though successive whorls do not overlap, this geometric form could not function as an *equivalved* bivalve because the two umbonal areas would conflict. Several taxonomic groups of bivalved organisms have developed means of avoiding this difficulty. One of these means is shown in text-figure 9. In this case, the axial margin of the whorl grows relatively more rapidly than the outer margin. Therefore, when two such valves are placed together along their growing edges the two umbos do not conflict and the space between them is sufficient for valve opening (see discussion of "interumbonal growth" by Stasek, 1963).

In the discussion thus far, the term generating curve has been used for a geometric figure lying in a plane containing the coiling axis. If, as in text-figure 8, the trace of the generating curve is extended it will pass through the axis of the coiling. In the bivalve shown in text-figure 9 the

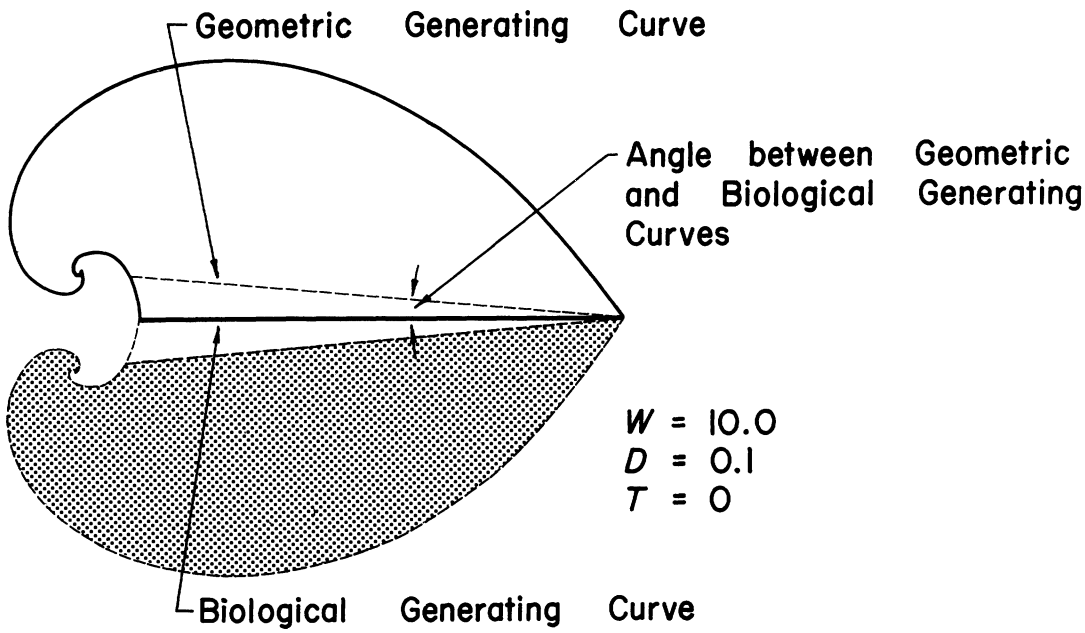


TEXT-FIG. 8—Hypothetical bivalve illustrating the necessity of interumbonal growth in equivalved bivalves.

generating curve does not coincide with the leading edge of the shell. Rather, we have what may be termed a *biological generating curve* which is at an angle to the *geometric generating curve*. The angle is large enough so that when two such shells are placed together the umbos do not interfere. It is evident, therefore, that equivalved bivalves are able to occupy the so-called bivalve region in text-figure 7 only because of a secondary adaptation: that of depositing a biological generating curve which is not coincident with the geometric generating curve.

The angle between the biological and the geometric generating curves, in combination with other geometric parameters, will determine the degree to which a bivalve can open. Several species of the pelecypod genera *Arca* and *Cardium* have umbos that have been beveled as a result of wear. The wear is caused by the fact that the organism attempted to open farther than the geometry allowed. In working with coiling in pelecypods the angle between the biological and geometric generating curves should be used as an additional basic parameter. Text-figure 10 shows several possible morphologies resulting only from variations in this angle.

There are other geometric mechanisms used by bivalved organisms that serve to increase potential for wide valve opening. A very simple example is the use of an increase in W to decrease the relative size of the umbo, thus providing more space between the umbos (as in *Arca senilis* as described by Yonge, 1955). Another mechanism involves pelecypods whose valves have different geometries. For example, the genera



TEXT-FIG. 9—Same shell form as in text-fig. 8 but with interumbonal growth (expressed by the angle between biological and geometric generating curves).

Exogyra and *Gryphaea* have one large valve and one small cap-like valve. Yet another class of solutions to the same problem is exemplified by the genus *Corculum*, as described by Bartsch (1947). Here, the two valves are approximately the same size but their geometries differ: the umbo of the left valve is displaced along the hinge line relative to the umbo of the right valve. When the shell opens, the two umbos move past each other.

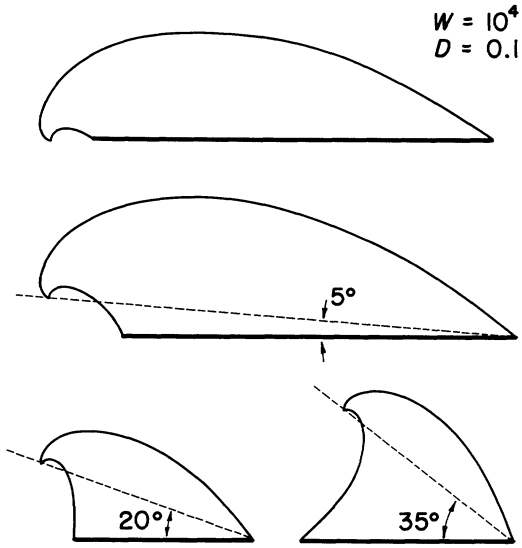
THE PROBLEM OF UNIVALVENESS

When considering the univalve shell we are relieved of the problem of requiring a functional hinge. Nevertheless, the univalve groups in text-figure 4 are confined to the upper part of the block and in general occupy a region not occupied by bivalves. At least one reason for this becomes apparent when we consider the primary functions of the shell in univalves.

The univalve shell serves in part for internal support of the soft part anatomy and in part for protection of the animal. The main function of the aperture is to provide access to the interior of the shell. The function of the shell is defeated if the size of the aperture becomes very large in relation to total shell volume. A large aperture inevitably reduces the *relative* amount of surface area insider the shell and detracts from the protective function of the shell. Returning to the basic relations shown in text-figures 2 and 4, we see that the ratio of apertural area to total shell volume varies systematically with changes in basic geometric parameters. It is thus reasonable that most univalve forms are confined to the upper regions of the three-dimensional block (i.e., relatively low expansion rate).

It should be noted further that the vast majority of univalve shells are above the surface which separates forms with overlapping whorls from those with separate whorls (text-fig. 7). In other words, univalves tend to have overlapping whorls. This is also reasonable geometrically. Where W is low (as in most univalves), the shell tends to be a tube which tapers rather gradually. If successive coils of this tube are not in contact, the strength of the coiled structure is lessened. Therefore, the presence of whorl overlap may be interpreted as serving to increase shell strength. This provides a second geometric reason for the concentration of univalves in the upper part of the block.

A few univalves depart radically from the foregoing generalizations. Limpets, for example, have large apertures. This is accomplished geometrically by a high expansion rate (W) and by a large angle between biological and geometric generating curves (see text-fig. 10). These facts do not nullify the reasoning applied above to typical univalves because the limpet shell type is used by organisms that live attached to a firm substratum and shell functions are not the same as for typical univalves. In a limpet, a large aperture serves to provide the necessary area for attachment of the foot to a substratum. As long as the animal remains attached, the protective function of aperture size is not a problem.



TEXT-FIG. *10*—Hypothetical shell forms illustrating the morphological effects of variation in the angle between biological and geometric generating curves. This angle is zero in the uppermost illustration.

An additional problem posed by the geometric relations of univalves is the region in text-figure 4 where both univalves and bivalves are found (gastropods and pelecypods, respectively). In this region are found shells that are quite bulky in relation to aperture size. The region is thus typically univalve. Following the logic used earlier, a truly equivalved bivalve employing this geometry would have difficulty functioning as a normal bivalve. The musculature would be insufficiently strong to wield the massive valves. Also, the shell would be poorly adapted for burrowing and, if epifaunal, would be very sensitive to the attempts of predators to open the shell (because the predator would be offered a scissors-like mechanism for opening the shell).

The fact is that the majority of bivalves found in this overlap region are forms that are strongly *inequivalved* (such as *Exogyra*). The smaller of the two valves is simply a cap. By reducing the mass of one valve the problem of musculature and its strength is alleviated and the scissors mechanism is avoided. Furthermore, the geometry of the whole shell (articulated valves) provides an excellent functional unit for the sessile

filter feeding organisms that generally occupy this region.

CONCLUSIONS

This analysis has been of a very general nature, being intended primarily as an introduction to a few general problems of interpreting coiling geometry and as an illustration of an approach to these problems. The basic model utilizes only four parameters and is obviously a simplification of the real situation. As soon as one attempts to consider some of the more subtle aspects of coiling variation, additional parameters are necessary (such as the angle between the biological and geometric generating curves). Regardless of the scale of the problem, however, the same type of logic may be applied.

Many questions remain to be answered. For example, why are brachiopods confined to the planispiral face of the three-dimensional block in text-figure 4 and why are pelecypods not found on this face? It is tempting to answer this question simply in terms of phylogeny: that is, to conclude that pelecypods have by chance never evolved the capability of truly planispiral coiling. To draw this conclusion is to discard the question and thereby to ignore the possibility of a rigorous functional explanation (perhaps in terms of musculature and hingement) for the coiling difference between the two phylogenetic groups.

Finally, it should be emphasized that the present study has not treated several important groups of coiled shells, particularly the nautiloids and foraminifera. The addition of these groups will be necessary for a complete understanding of the coiling phenomenon.

REFERENCES

BARTSCH, P., 1947, The little hearts (*Corculum*) of the Pacific and Indian Oceans: Pacific Sci., v. 1, p. 221–226.

FUKUTOMI, T., 1953, A general equation indicating the regular forms of Mollusca shells, and its application to geology, especially in paleontology (I): Hokkaido Univ., Geophys. Bull., v. 3, p. 63–82.

GRABAU, A. H., 1872, Über due Naumann'sche Conchospirale und ihre Bedeutung für die Conchyliometrie: Univ. Leipzig, Inauguraldissert., Leipzig, 1872, s. 1–77.

LISON, L., 1949, Recherches sur la forme et la mechanique de développement des coquilles des lamellibranches: Inst. Roy. Sci. Nat. Belg., Mém., fasc. 34, p. 1–87.

MOSELEY, H., 1838, On the geometrical forms of turbinated and discoid shells: Roy. Soc. Lon., Philos. Trans. for 1838, p. 351–370.

——, 1842, On conchyliometry: Philos. Mag., v. 21, p. 300–305.

MÜLLER, J. H. T., 1850, Beiträge zur Conchyliometrie: Ann. der Phys., Bd. 81, s. 533–544.

——, 1853, Zweiter Beitrag zur Conchyliometrie: *op. cit.*, Bd. 90. s. 323–327.

NAUMANN, C. F., 1840a, Beitrag zur Conchyliometrie: *op. cit.*, Bd. 50, s. 223–236.

——, 1840b, Über die Spiralen der Ammoniten: *op. cit.*, Bd. 51, s. 245–259.

——, 1845, Über die wahre Spirale der Ammoniten: *op. cit.*, Bd. 64, s. 538–543.

——, 1848, Über die logarithmische Spiral von *Nautilus pompilius* und *Ammonites galeatus:* Ber. Verh. k. sächs. Gesell. Wissen., Bd. 2, s. 26–34.

OWEN, G., 1953, The shell in the Lamellibranchia: Quart. Jour. Micros. Sci., v. 94, p. 57–70.

RAUP, D. M., 1961, The geometry of coiling in gastropods: Nat. Acad. Sci., Proc., v. 47, p. 602–609.

——, 1962, Computer as aid in describing form in gastropod shells: Science, v. 138, p. 150–152.

——, 1963, Analysis of shell form in gastropods: Geol. Soc. Amer., Spec. Pap. 73, p. 222.

——, in press, Geometric analysis of shell coiling: Coiling in ammonoids: Jour. Paleontology, v. 41.

—— & A. Michelson, 1965, Theoretical morphology of the coiled shell: Science, v. 147, p. 1294–1295.

RUDWICK, M. J. S., 1959, The growth and form of brachiopod shells: Geol. Mag., v. 96. p. 1–24.

STASEK, C. R., 1963, Geometrical form and gnomonic growth in bivalved Mollusca: Jour. Morph., v. 112, p. 213–231.

THOMPSON, D'A. W., 1942, On Growth and Form. Camb. Univ. Press, 1116 p.

YONGE, C. M., 1955, A note on *Arca* (*Senilia*) *senilis* Lamarck: Malac. Soc. Lon., Proc., v. 31, p. 202–208.

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